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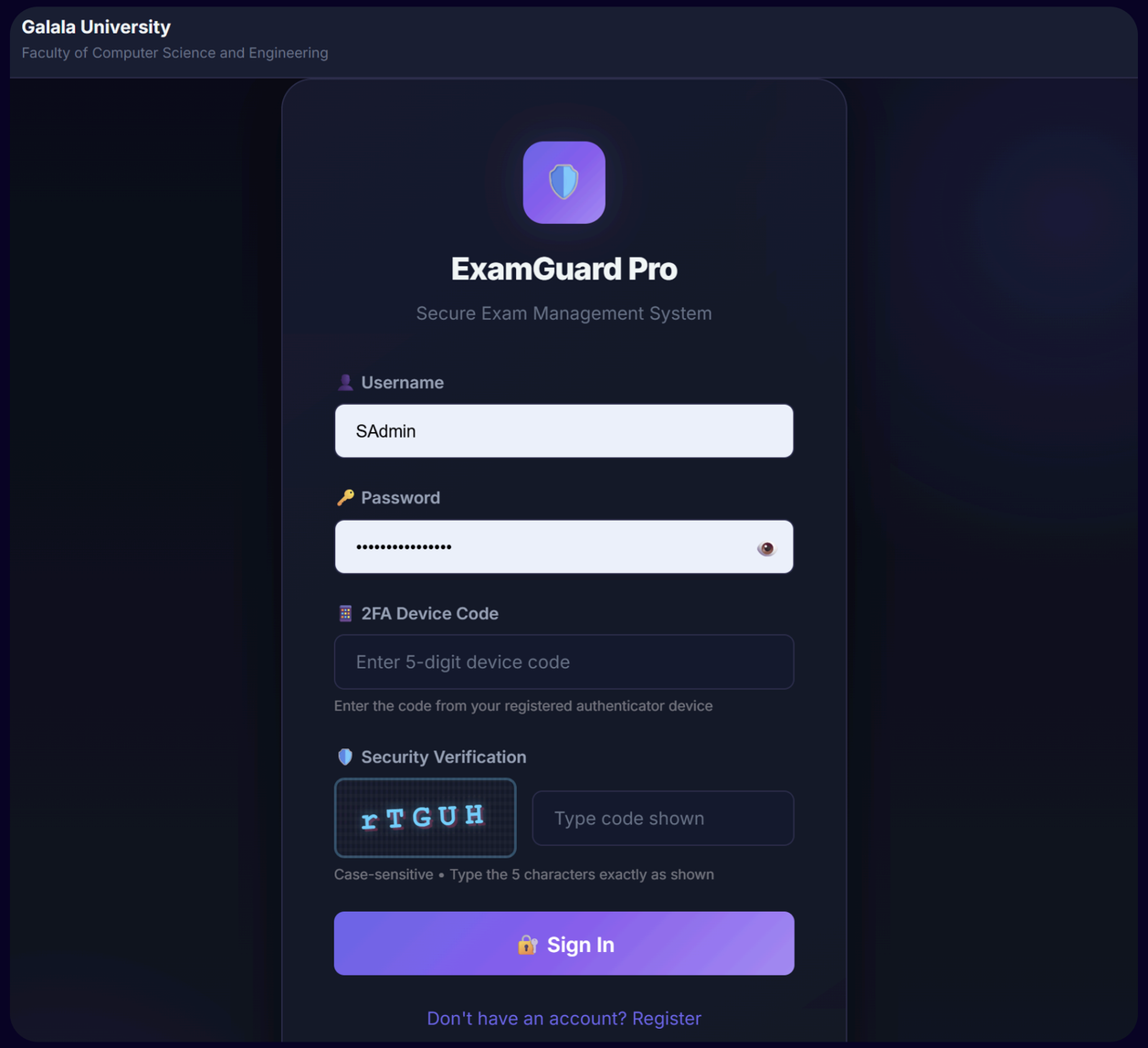
EXAM MANAGEMENT SYSTEM



CSE251 Software Engineering Course Project
Supervised By: Dr/ Nora Niazy
Spring 2025/26

Team:
Omar Khalifa 223101614 CE
Samaa Mohamed 223101629 CE

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OVERVIEW

ExamGuard is a full-stack web-based Exam Management System designed to digitize the complete examination lifecycle from question authoring to grading and analytics. It enforces strict role separation and applies a security-first design at every layer.

GOALS

- Model system with 8 UML diagrams
- Implement a full-stack working prototype
- Enforce RBAC/ABAC + encryption throughout
- Apply STRIDE threat modeling
- Validate all requirements through structured testing

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SYSTEM INTERFACES AND USERS

Student

Students who take exams in strict modes. Chat with teachers, Enter Groups or take roles for Projetcs, Join Exam rooms and review their grades

Super Admin

a Super Admin who manages the platform. They can Edit users and roles and accessability. Read Logs and Monitor The platform

Teacher

a Teacher who creates and grades exams, Can assign groups or students roles, chat with students.



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FUNCTIONAL REQUIREMENTS:

functional requirements were gathered during requirements analysis, covering authentication, exam lifecycle, grading, communication, and administration.

ID	Requirement
FR-01	Users can register with validated credentials (username, email, password)
FR-02	Users can log in with CAPTCHA and 2FA verification
FR-03	Instructors can create, edit, publish, and delete exams
FR-04	Exams support MCQ, True/False, and open-ended question types
FR-05	Students can take published exams within configured time windows
FR-06	System auto-grades MCQ/True-False and calculates scores
FR-07	Admin manages users, roles, groups, and system configuration
FR-08	Encrypted real-time chat between teachers and students
FR-09	Dashboard displays performance analytics and exam statistics

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NON-FUNCTIONAL REQUIREMENTS:

Category	Requirement
Security	AES-256 encryption, RBAC/ABAC, CSRF protection, rate limiting
Performance	Page load < 2s, support 100+ concurrent exam sessions
Usability	Responsive UI, intuitive navigation, role-specific dashboards
Reliability	Auto-save exam answers, session recovery, data backup
Scalability	Modular architecture, schema portable to PostgreSQL
Auditability	Immutable append-only audit logs for all security events

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SYSTEM ARCHITECTURE & TECHNOLOGY

Frontend
HTML5 + CSS3 + Vanilla JS
EJS Templates

Backend
Node.js + Express.js
RESTful API Routes

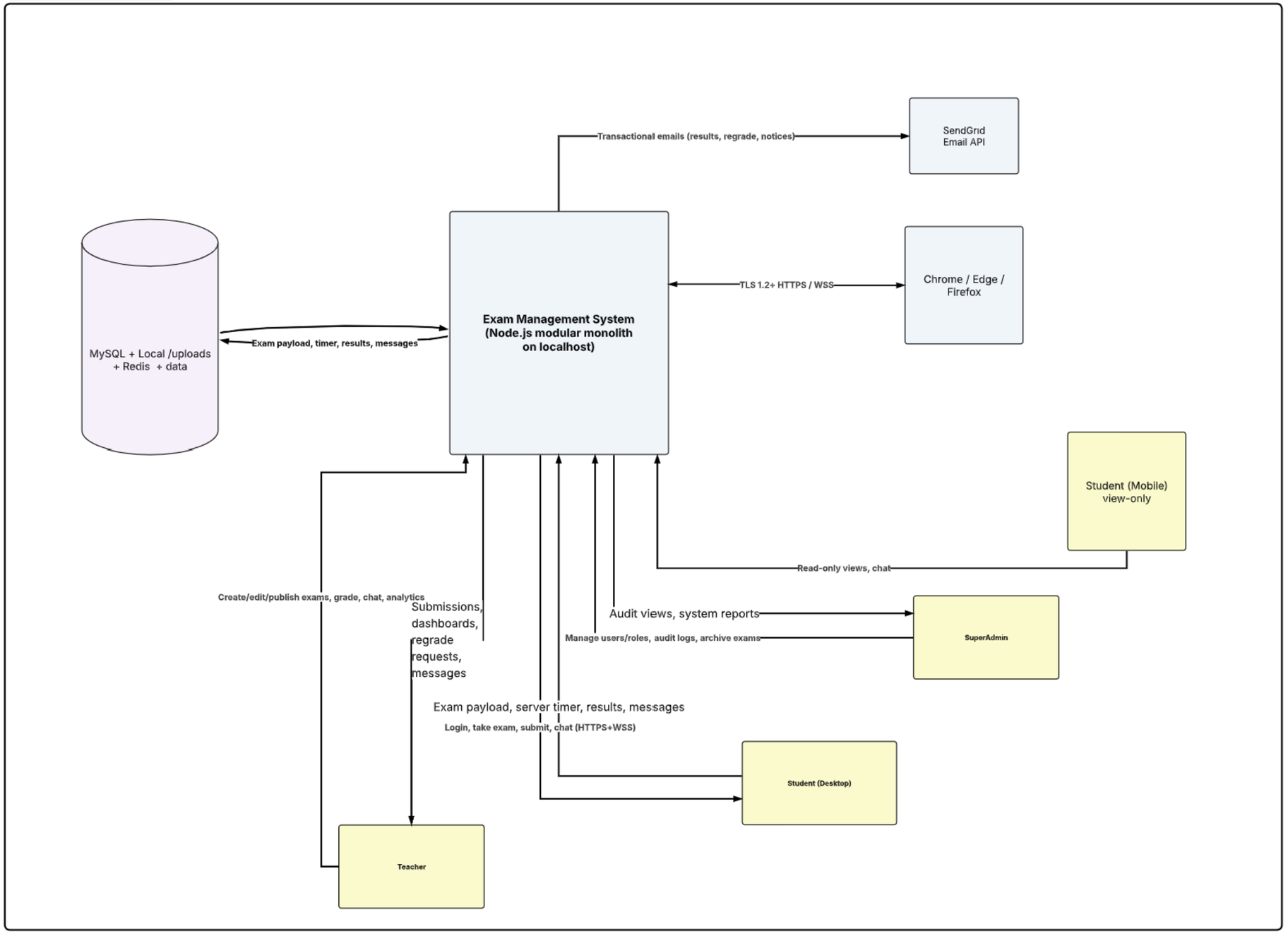
Middleware
Auth · RBAC · ABAC · CSRF
Rate Limiter · Audit Log · Validator

Database
SQLite3 (better-sqlite3)
12 Tables · Parameterized Queries

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CONTEXT DIAGRAM

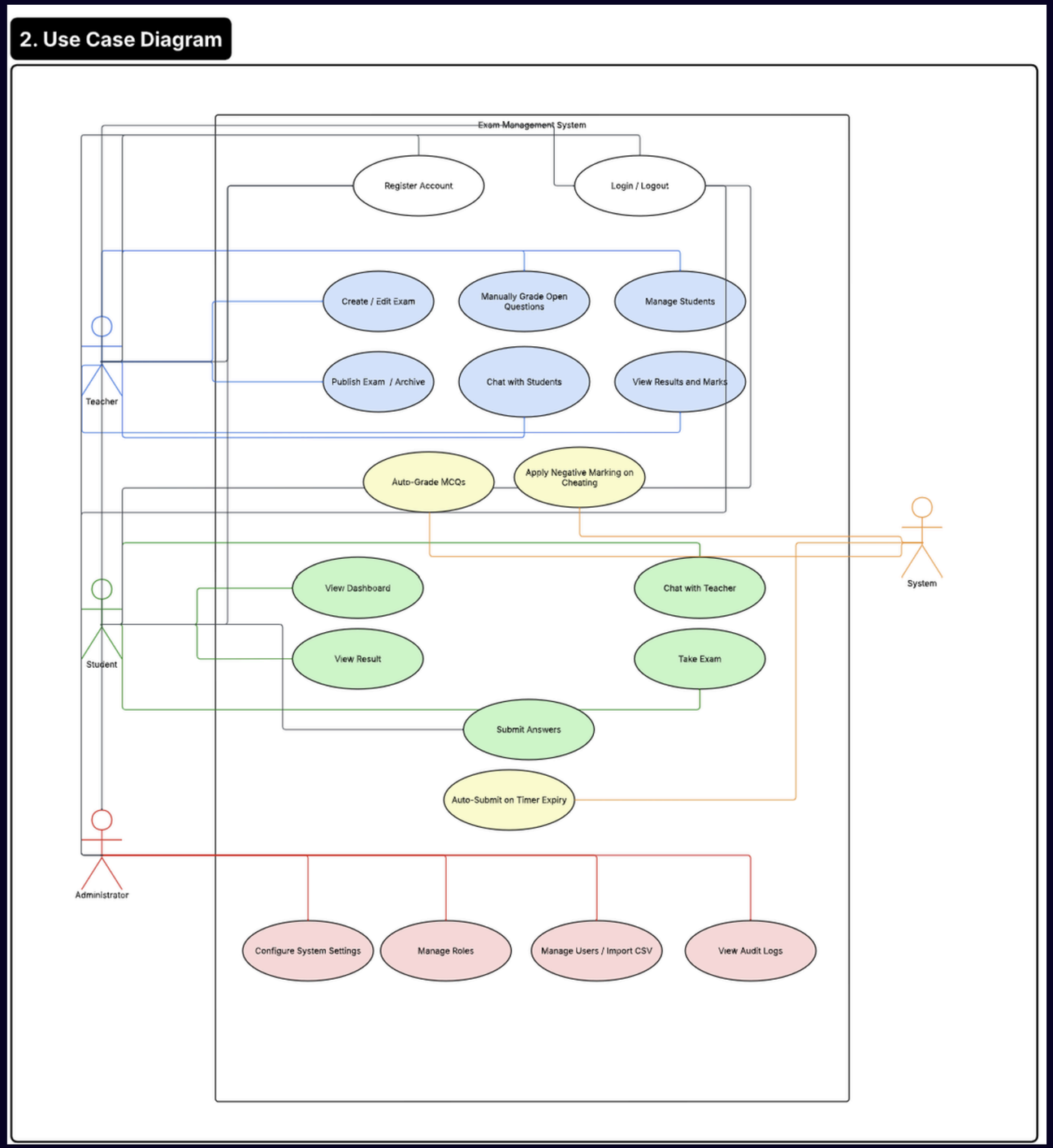
1. Context Diagram



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USE CASE DIAGRAM

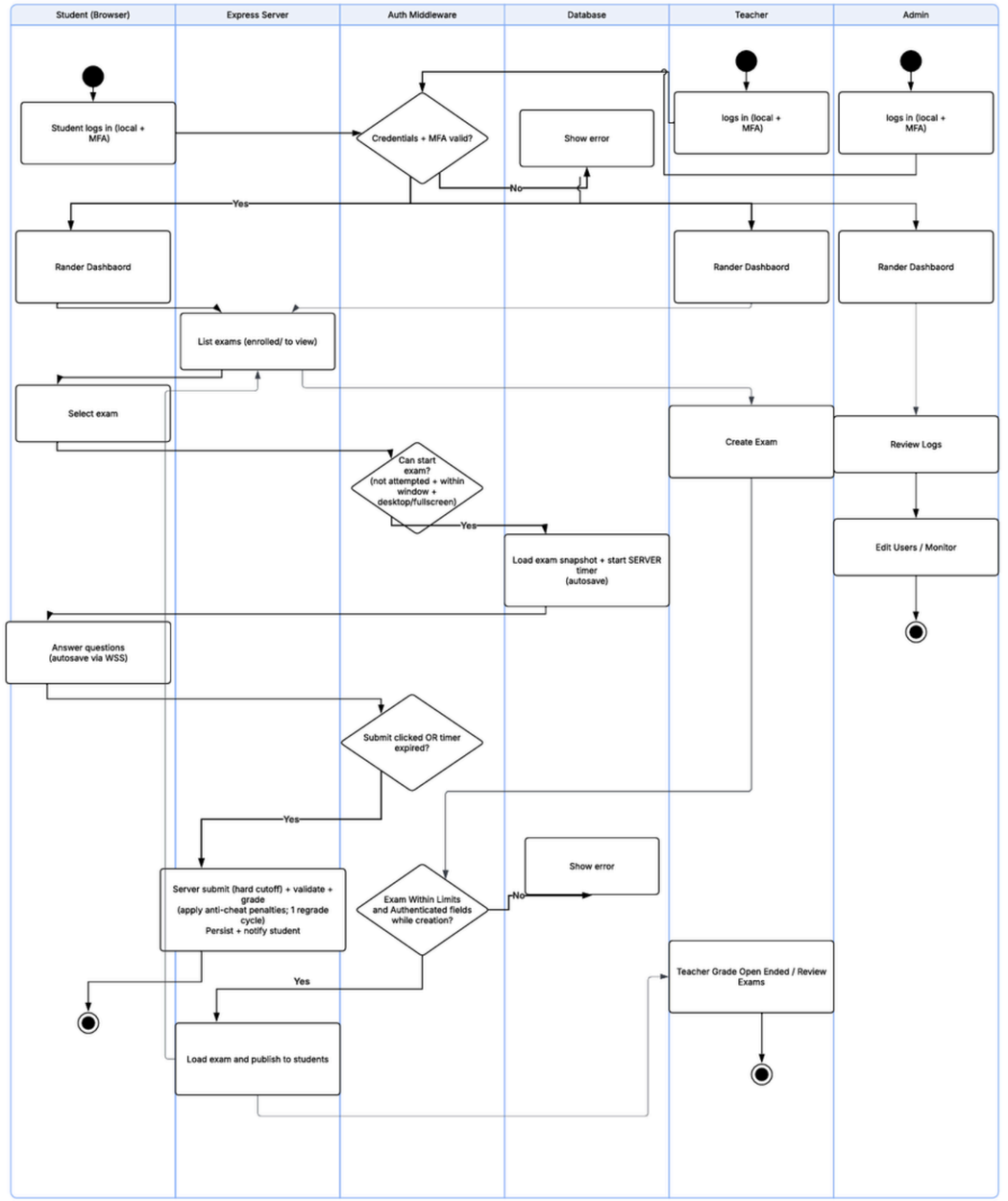
2. Use Case Diagram



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ACTIVITY DIAGRAM

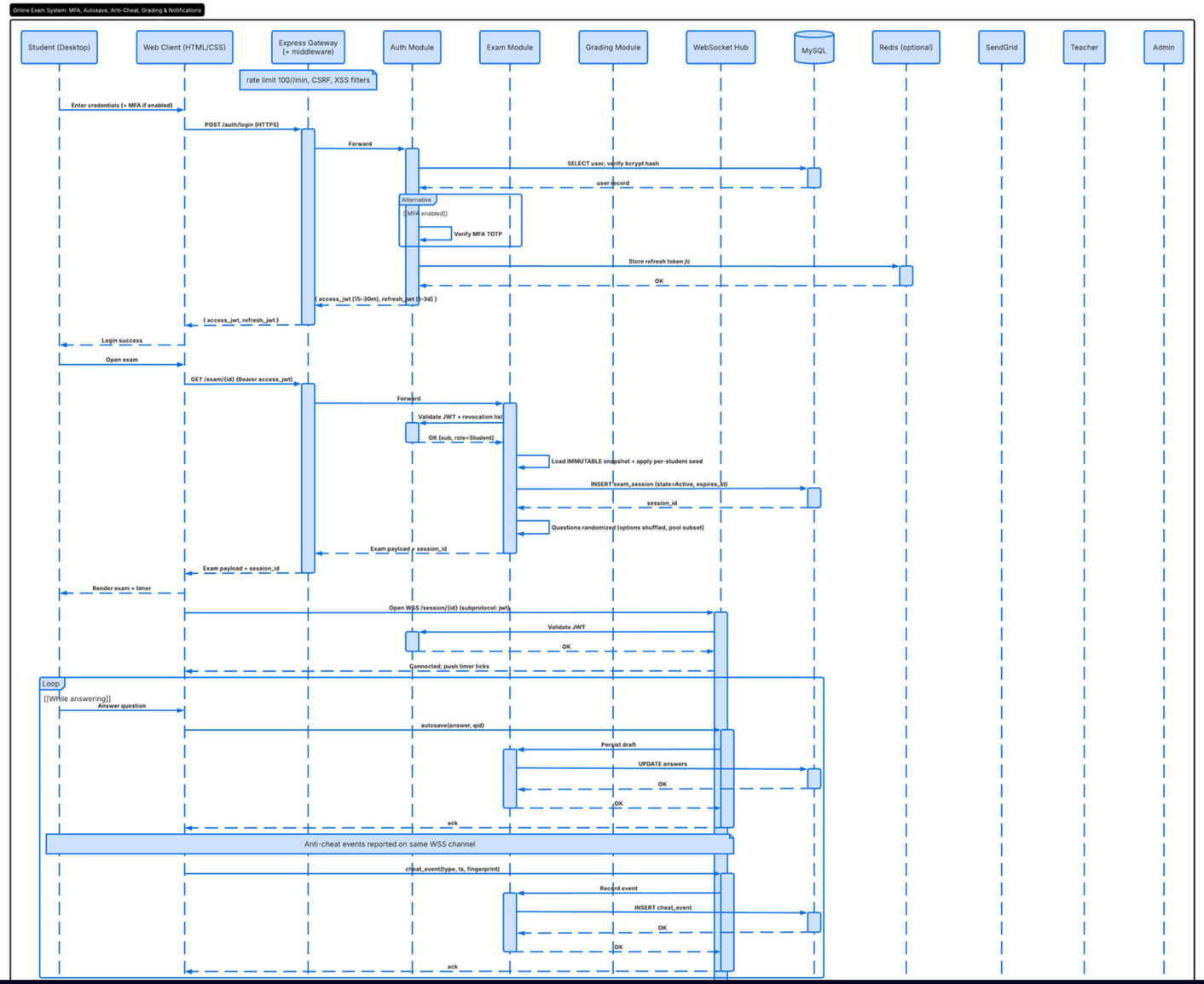
3. Activity Diagram



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SEQUENCE DIAGRAM

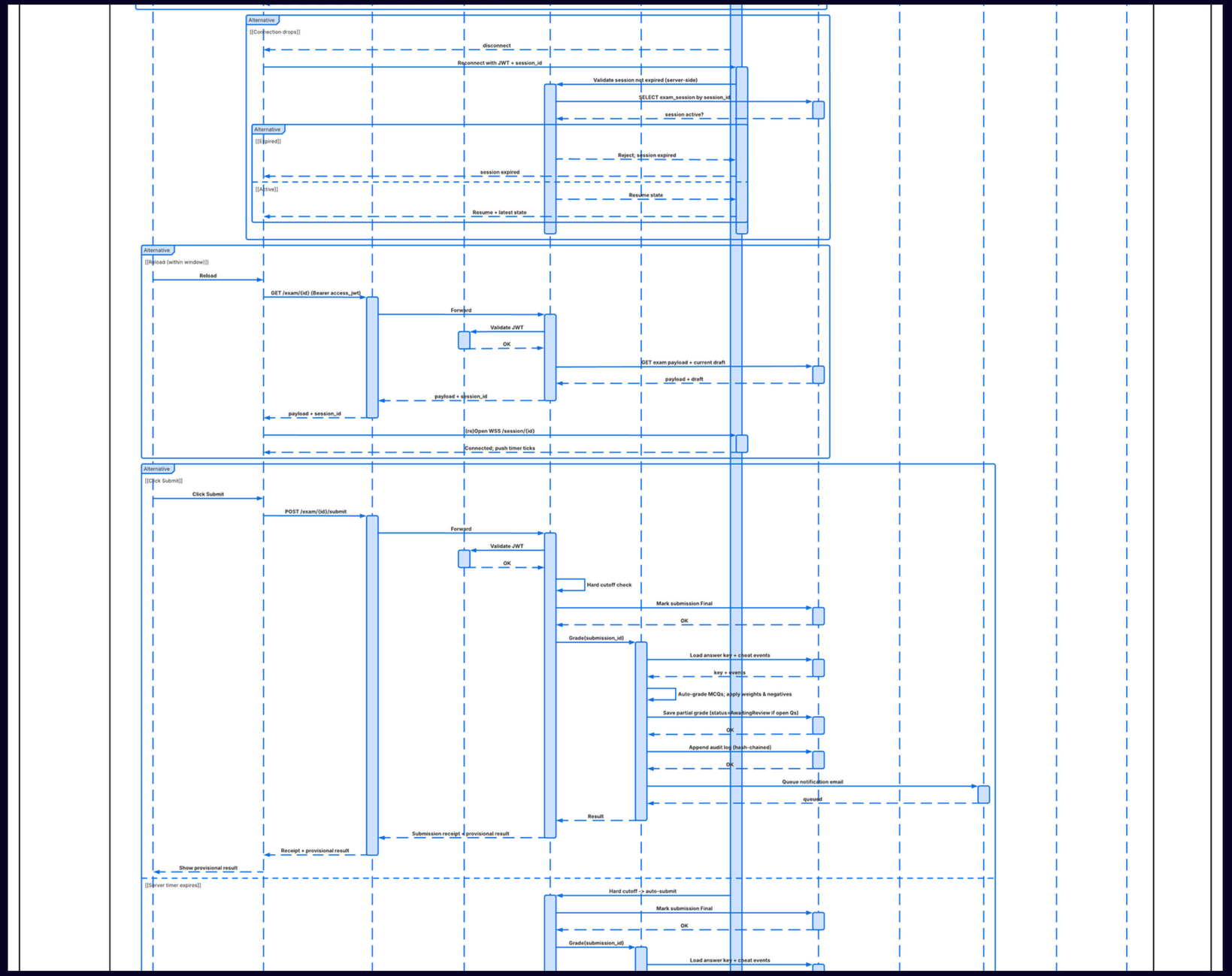
4. Sequence Diagram



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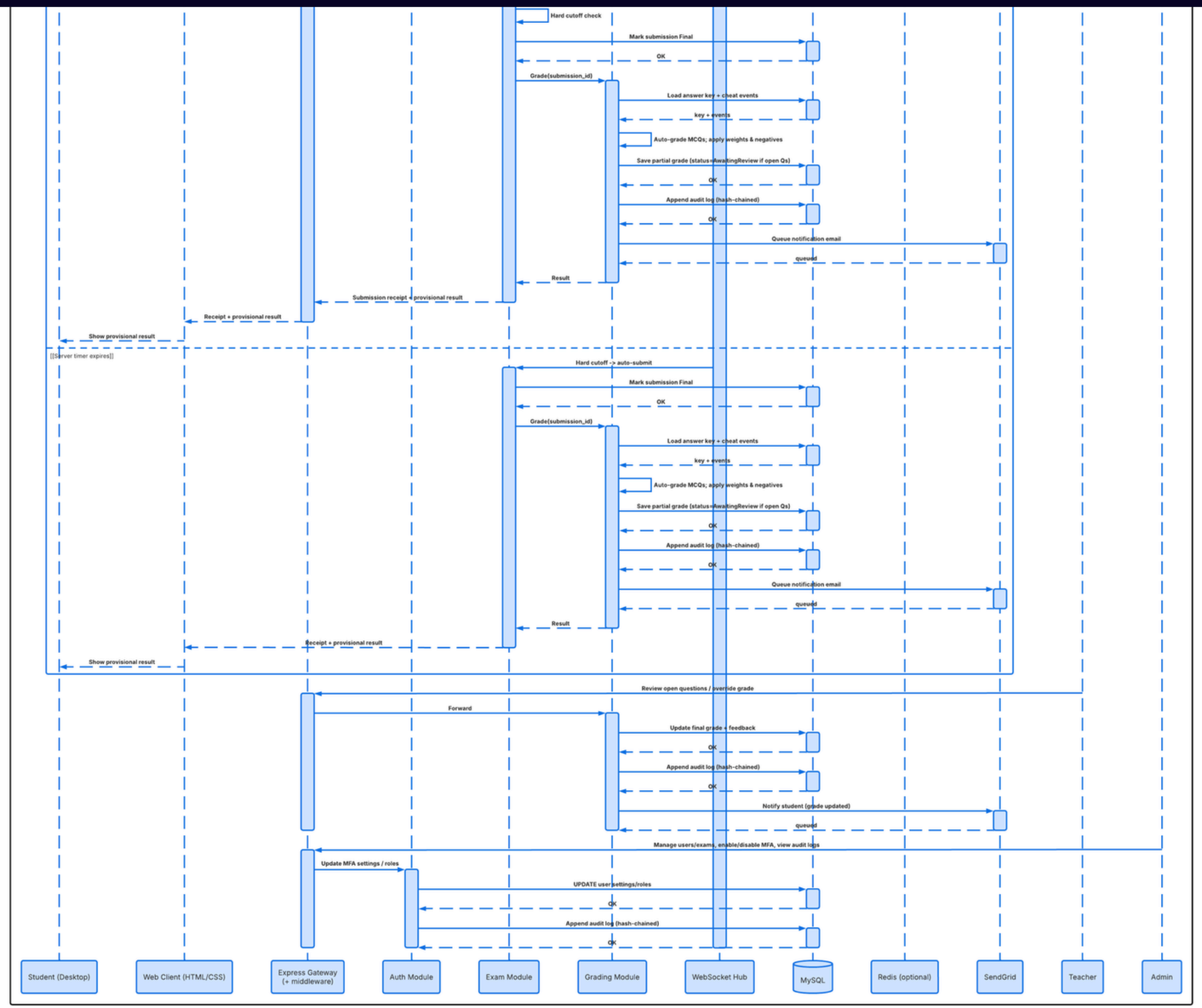
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SEQUENCE DIAGRAM



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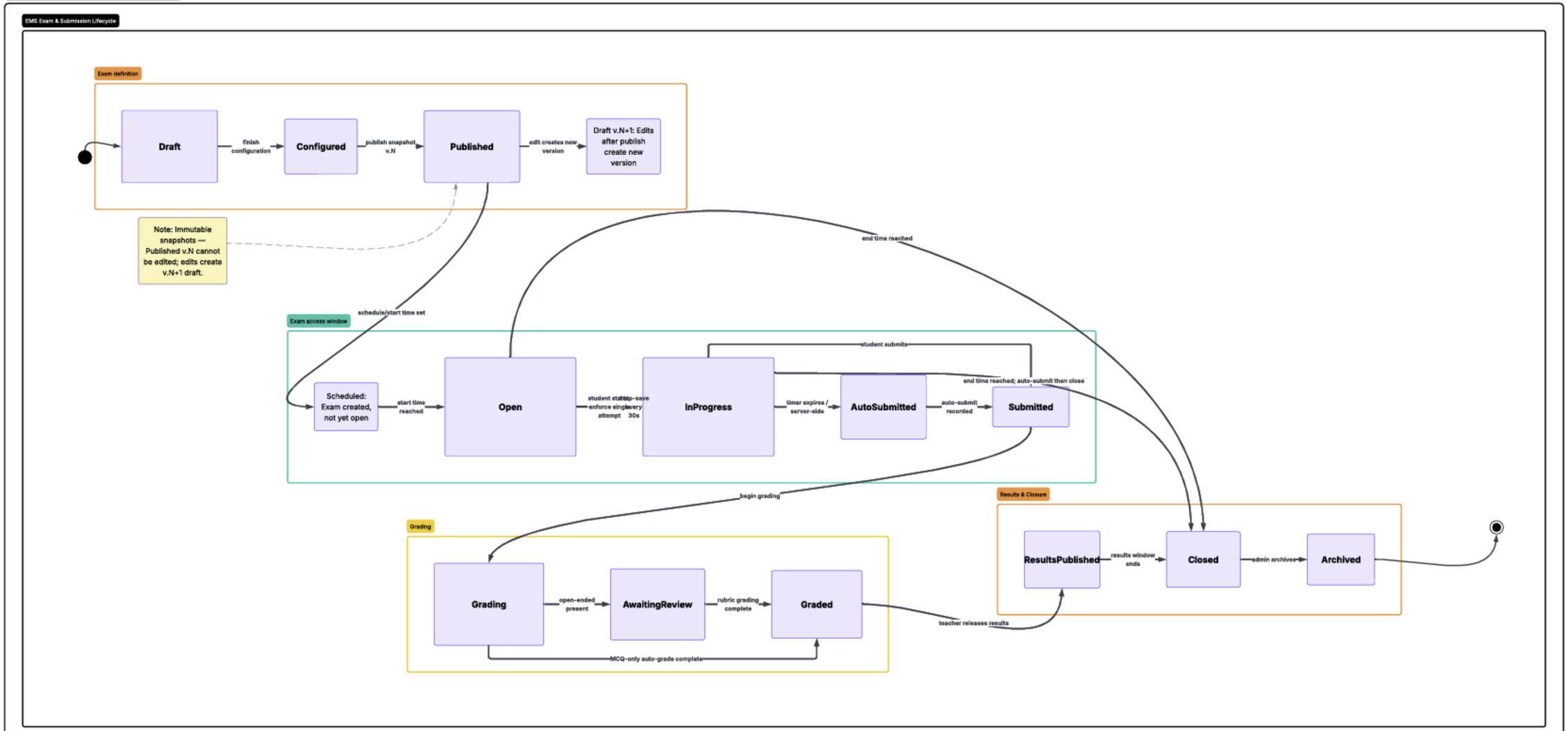
SEQUENCE DIAGRAM



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STATE DIAGRAM

5. State Machine Diagram

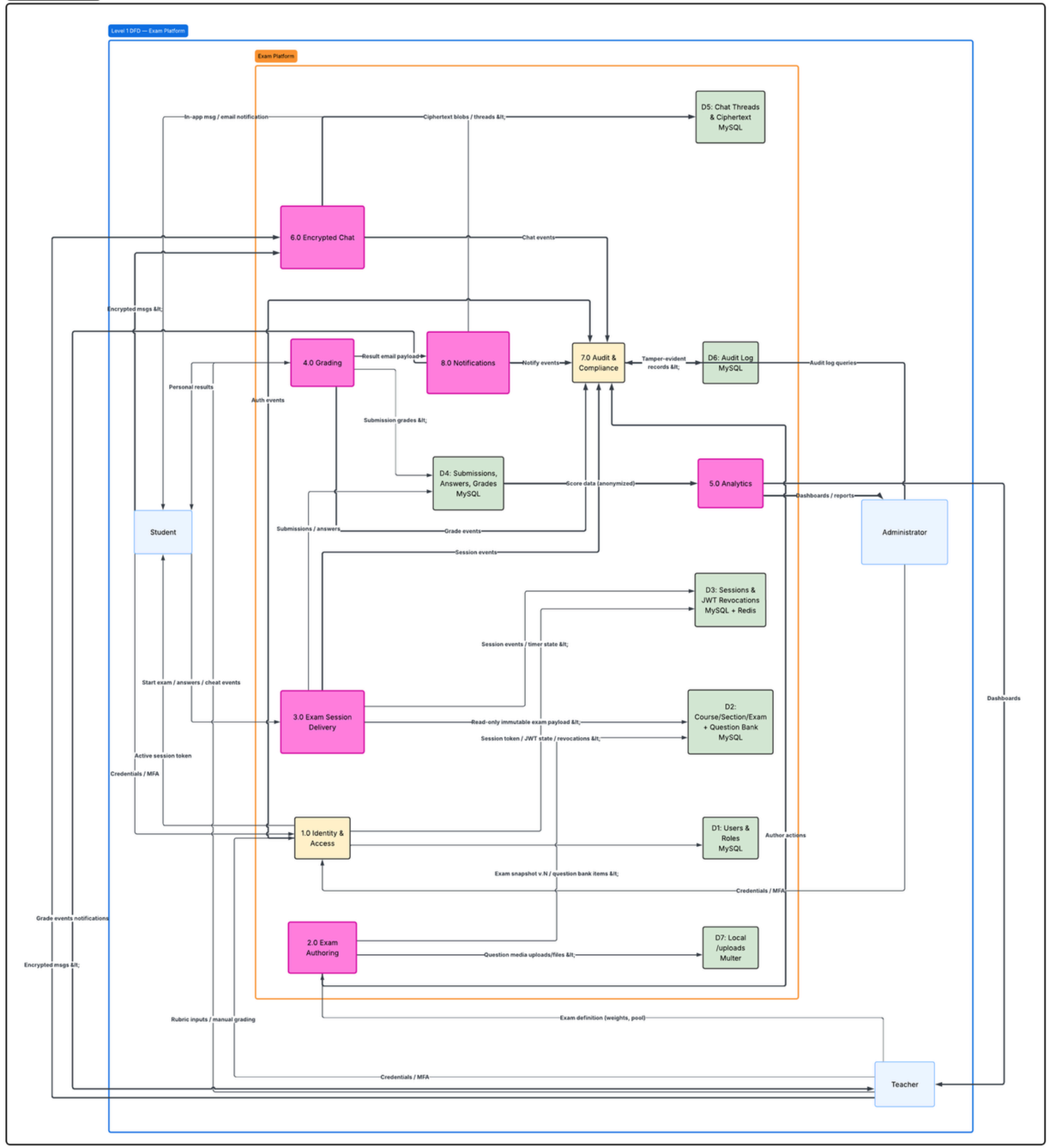


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DFD

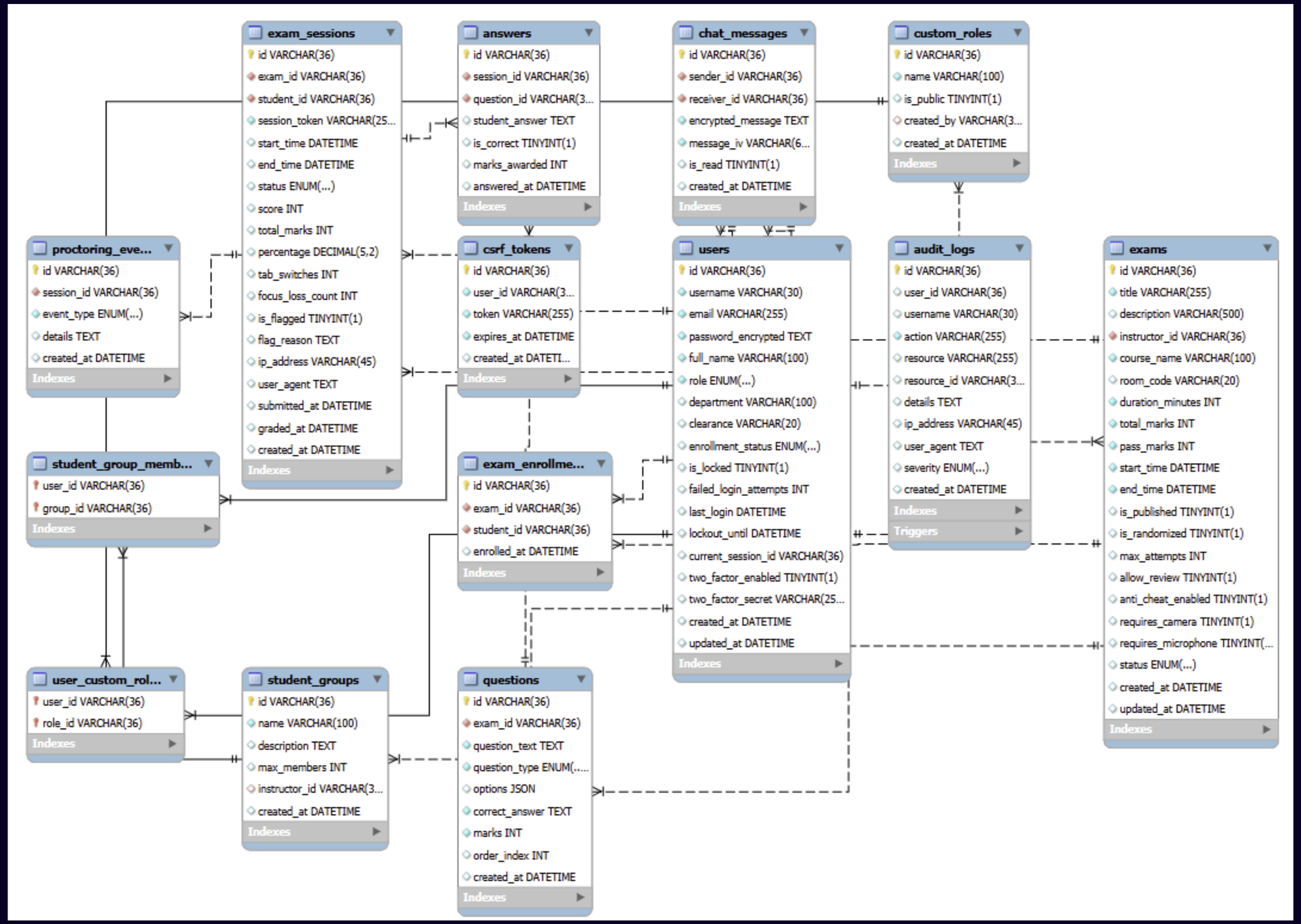
6. DFD Level 1



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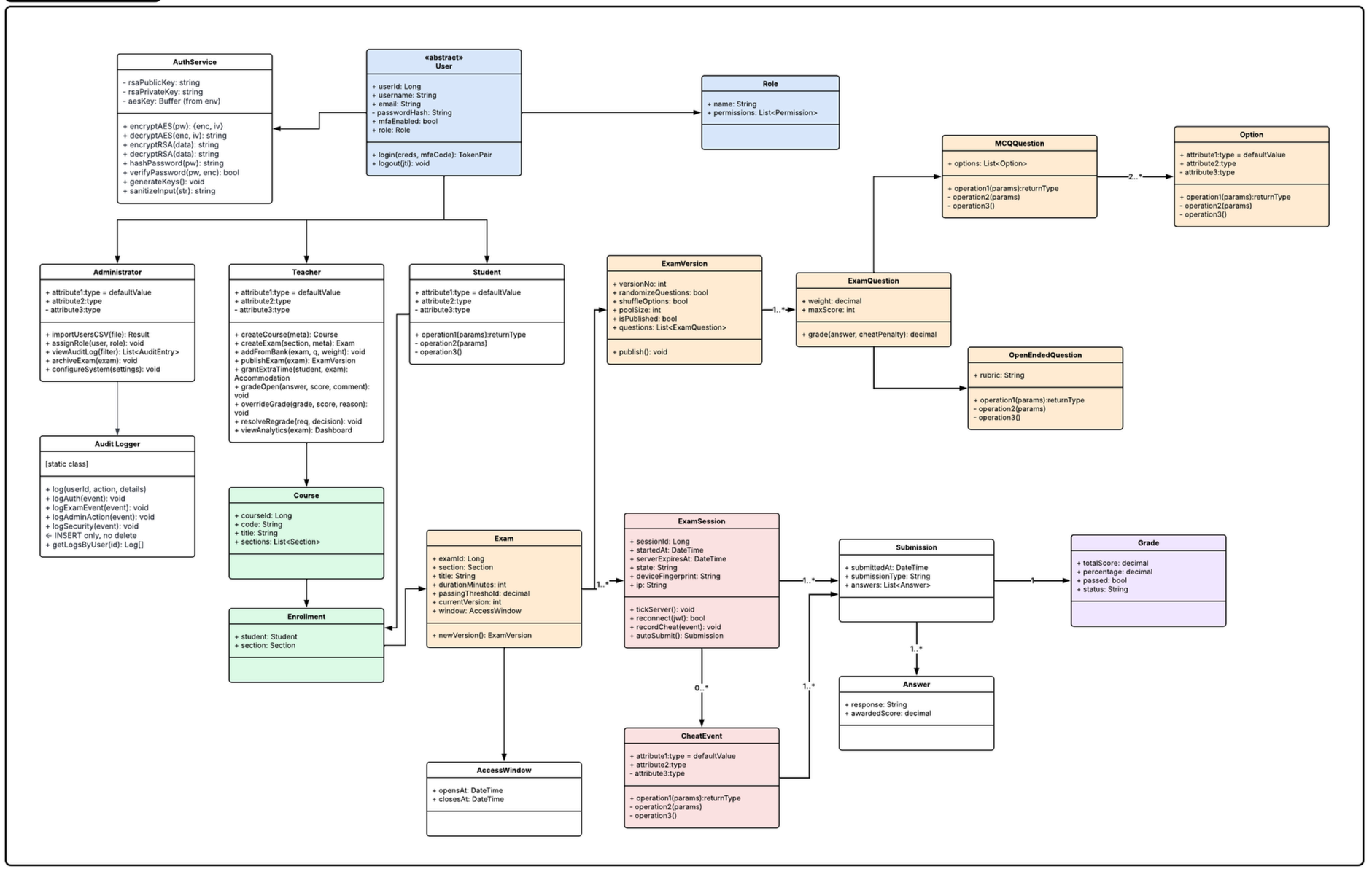
RELATIONAL ERD



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CLASS DIAGRAM

8. Class Diagram



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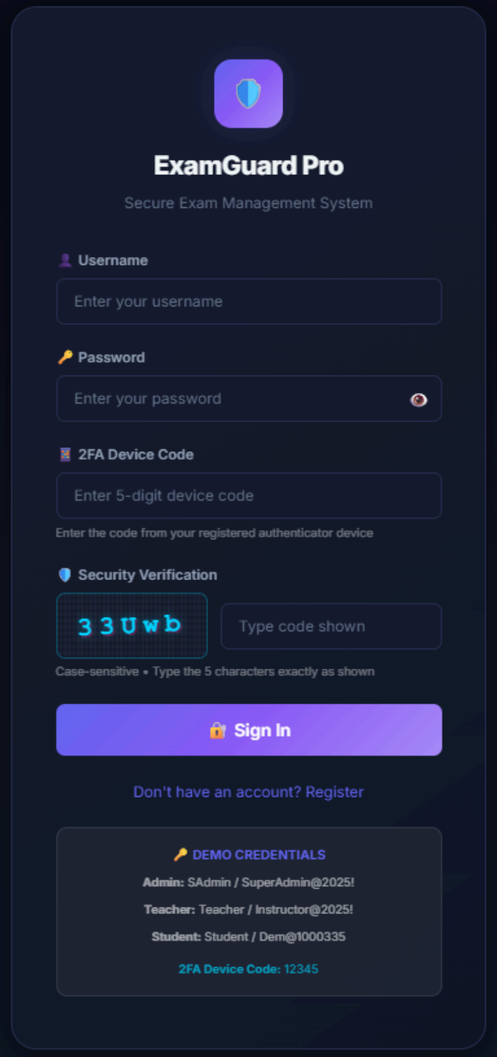
STRIDE THREAT MODEL

Threat	Description	Mitigation
Spoofing	Fake identity / impersonation	JWT auth, CAPTCHA, 2FA, session binding
Tampering	Modify data in transit/storage	CSRF tokens, input validation, parameterized SQL
Repudiation	Deny performing an action	Immutable append-only audit log, IP tracking
Info Disclosure	Expose sensitive data	AES-256-CBC, PBKDF2, RSA, generic errors
Denial of Service	Overwhelm system resources	Rate limiting (100 req/min), account lockout
Elevation of Privilege	Gain unauthorized access	RBAC hierarchy + ABAC (dept+time+clearance)

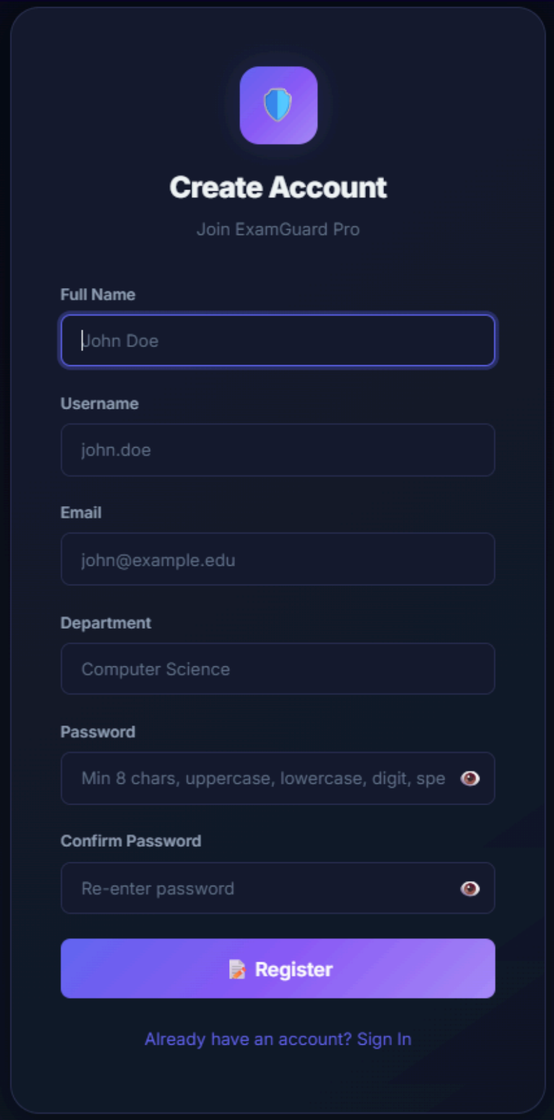
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DEMO TIME

LOG-IN



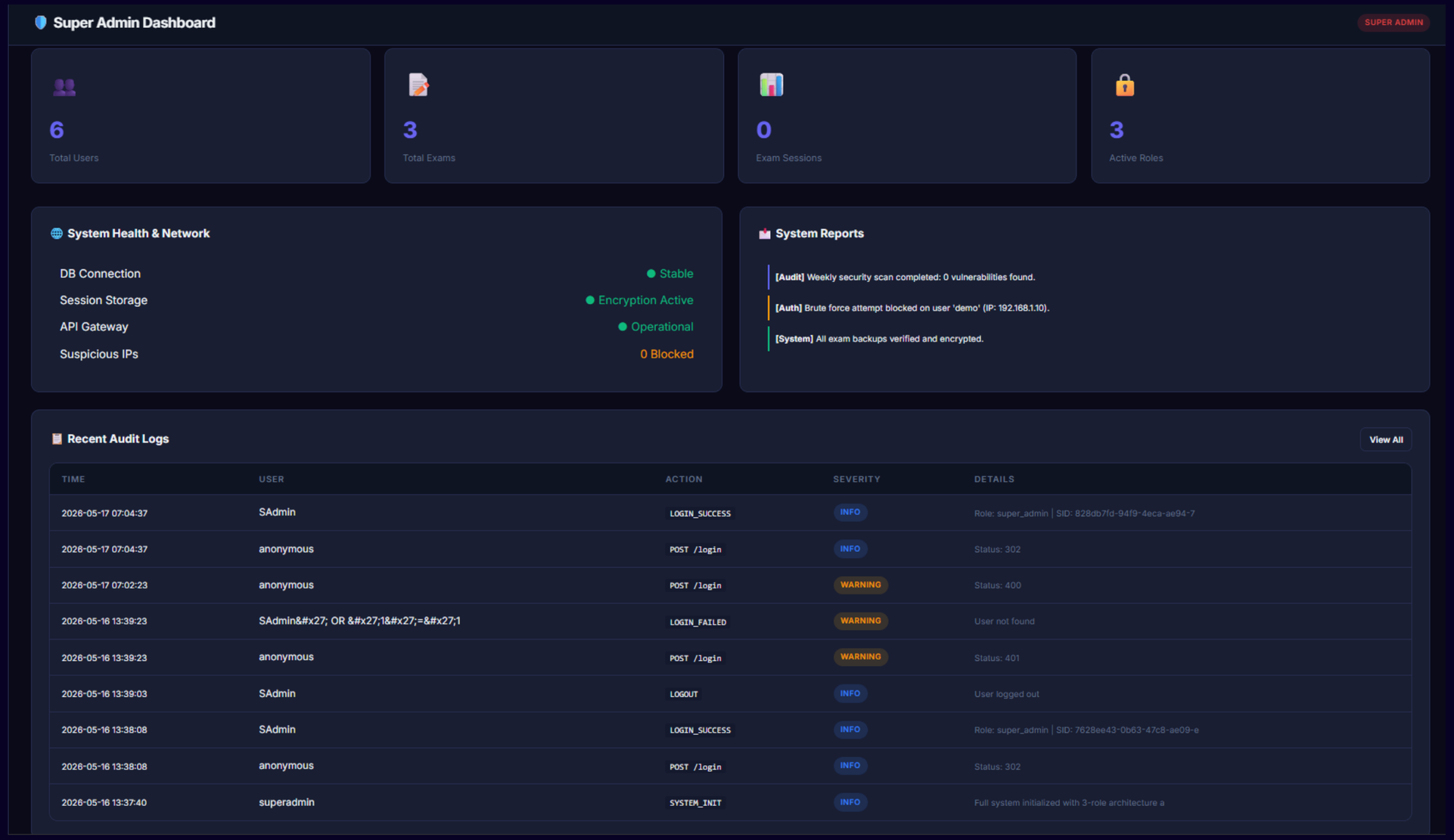
SIGN-UP



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ADMIN INTERFACE

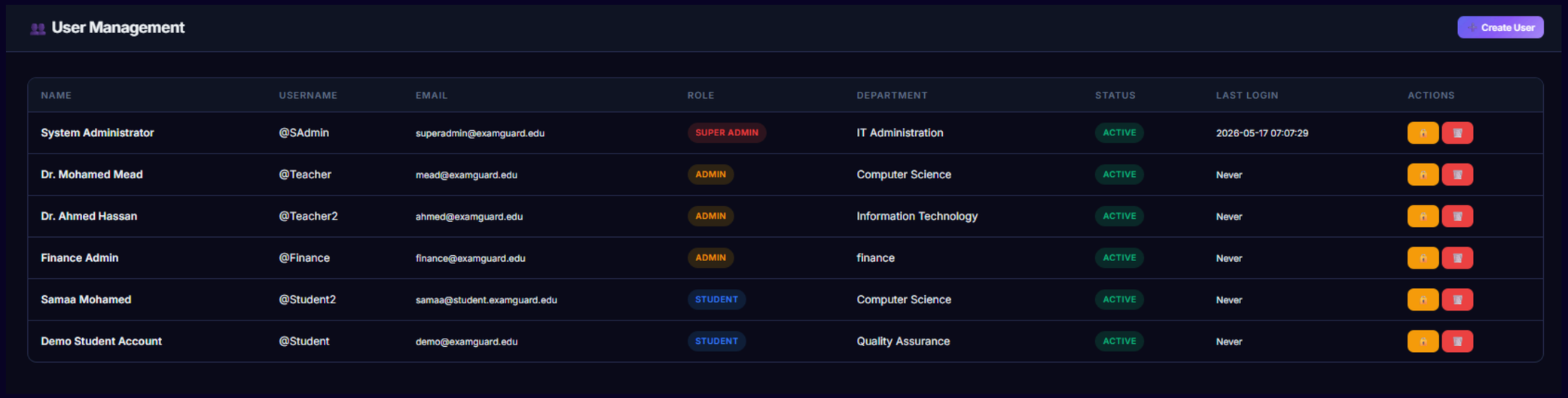
ADMIN DASHBOARD



PEOPLE



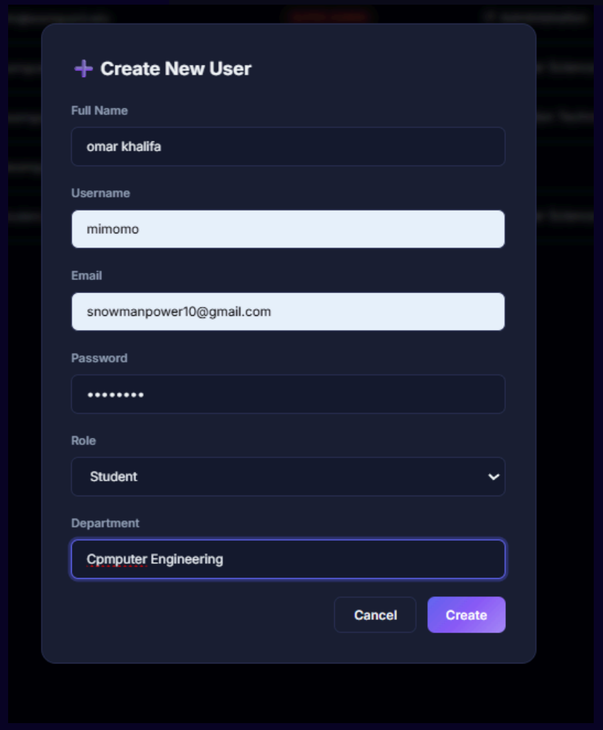
USER MANAGEMENT



REMOVING A USER

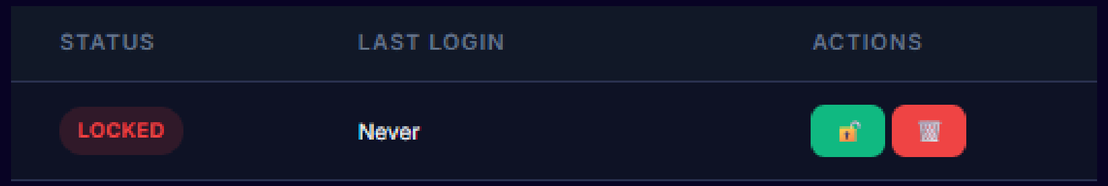
NAME	USERNAME	EMAIL	ROLE	DEPARTMENT	STATUS	LAST LOGIN	ACTIONS
System Administrator	@SAdmin	superadmin@examguard.edu	SUPER ADMIN	IT Administration	ACTIVE	2026-05-17 07:07:29	
Dr. Mohamed Mead	@Teacher	mead@examguard.edu	ADMIN	Computer Science	ACTIVE	Never	
Dr. Ahmed Hassan	@Teacher2	ahmed@examguard.edu	ADMIN	Information Technology	ACTIVE	Never	
Finance Admin	@Finance	finance@examguard.edu	ADMIN	finance	ACTIVE	Never	
Samaa Mohamed	@Student2	samaa@student.examguard.edu	STUDENT	Computer Science	ACTIVE	Never	

ADDING A NEW USER



NAME	USERNAME	EMAIL	ROLE	DEPARTMENT	STATUS	LAST LOGIN
omar khalifa	@mimomo	snowmanpower10@gmail.com	STUDENT	Cpmputer Engineering	ACTIVE	Never
System Administrator	@SAdmin	superadmin@examguard.edu	SUPER ADMIN	IT Administration	ACTIVE	2026-05-17 07:07:29
Dr. Mohamed Mead	@Teacher	mead@examguard.edu	ADMIN	Computer Science	ACTIVE	Never
Dr. Ahmed Hassan	@Teacher2	ahmed@examguard.edu	ADMIN	Information Technology	ACTIVE	Never
Finance Admin	@Finance	finance@examguard.edu	ADMIN	finance	ACTIVE	Never
Samaa Mohamed	@Student2	samaa@student.examguard.edu	STUDENT	Computer Science	ACTIVE	Never

LOCKING A USER ACCOUNT



AUDIT LOGS

Audit Logs

32 ENTRIES

Severity: All

Action: Filter by action...

Filter

TIMESTAMP	USER	ACTION	RESOURCE	SEVERITY	IP	DETAILS
5/17/2026 07:12 AM	SAdmin	USER_LOCKED	user	WARNING	::1	mimomo
5/17/2026 07:12 AM	SAdmin	POST /users/d4ab1331-780a-42b0-bf39-b8ecdee6c97a/toggle-lock	/users/d4ab1331-780a-42b0-bf39-b8ecdee6c97a/toggle-lock	INFO	::1	Status: 302
5/17/2026 07:12 AM	SAdmin	USER_UNLOCKED	user	WARNING	::1	mimomo
5/17/2026 07:12 AM	SAdmin	POST /users/d4ab1331-780a-42b0-bf39-b8ecdee6c97a/toggle-lock	/users/d4ab1331-780a-42b0-bf39-b8ecdee6c97a/toggle-lock	INFO	::1	Status: 302
5/17/2026 07:12 AM	SAdmin	USER_LOCKED	user	WARNING	::1	mimomo
5/17/2026 07:12 AM	SAdmin	POST /users/d4ab1331-780a-42b0-bf39-b8ecdee6c97a/toggle-lock	/users/d4ab1331-780a-42b0-bf39-b8ecdee6c97a/toggle-lock	INFO	::1	Status: 302
5/17/2026 07:11 AM	SAdmin	USER_CREATED	user	INFO	::1	Created student: mimomo
5/17/2026 07:11 AM	SAdmin	POST /users/create	/users/create	INFO	::1	Status: 302
5/17/2026 07:10 AM	SAdmin	USER_DELETED	user	CRITICAL	::1	mimomo
5/17/2026	SAdmin	POST /users/c0ce2e70-2010-4c85-9a57-429df58b638b/delete	/users/c0ce2e70-2010-4c85-9a57-429df58b638b/delete	INFO	::1	Status: 302

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NETWORK STATUS

Network & System Status

SYSTEM

3h 38m
System Uptime

3182 MB
Free Memory (of 16203 MB)

Active
Security Layer Status

Network Interfaces

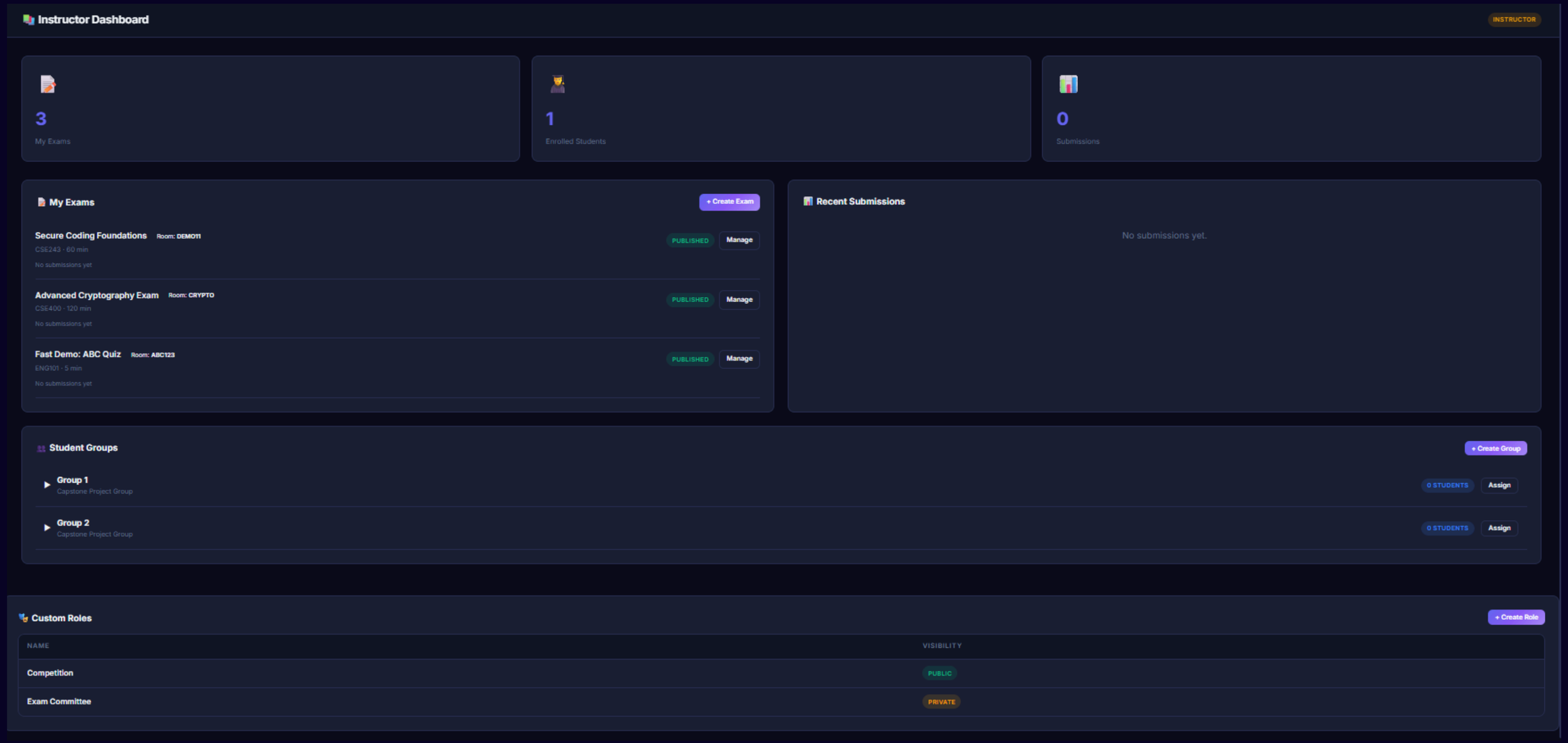
INTERFACE NAME	IP ADDRESS	MAC ADDRESS	FAMILY	INTERNAL
VMware Network Adapter VMnet1	169.254.176.49	00:50:56:c0:00:01	IPV4	NO
VMware Network Adapter VMnet8	169.254.228.30	00:50:56:c0:00:08	IPV4	NO
Wi-Fi	fe80::1afa:cee7:3bce:dfbc	fa:fb:17:f2:23:08	IPV6	NO
Wi-Fi	10.180.24.42	fa:fb:17:f2:23:08	IPV4	NO
Loopback Pseudo-Interface 1	::1	00:00:00:00:00:00	IPV6	YES
Loopback Pseudo-Interface 1	127.0.0.1	00:00:00:00:00:00	IPV4	YES

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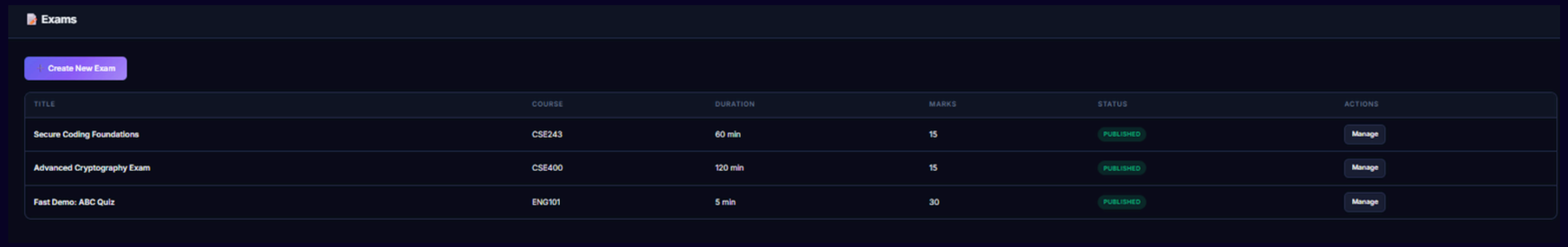
INSTRUCTOR INTERFACE

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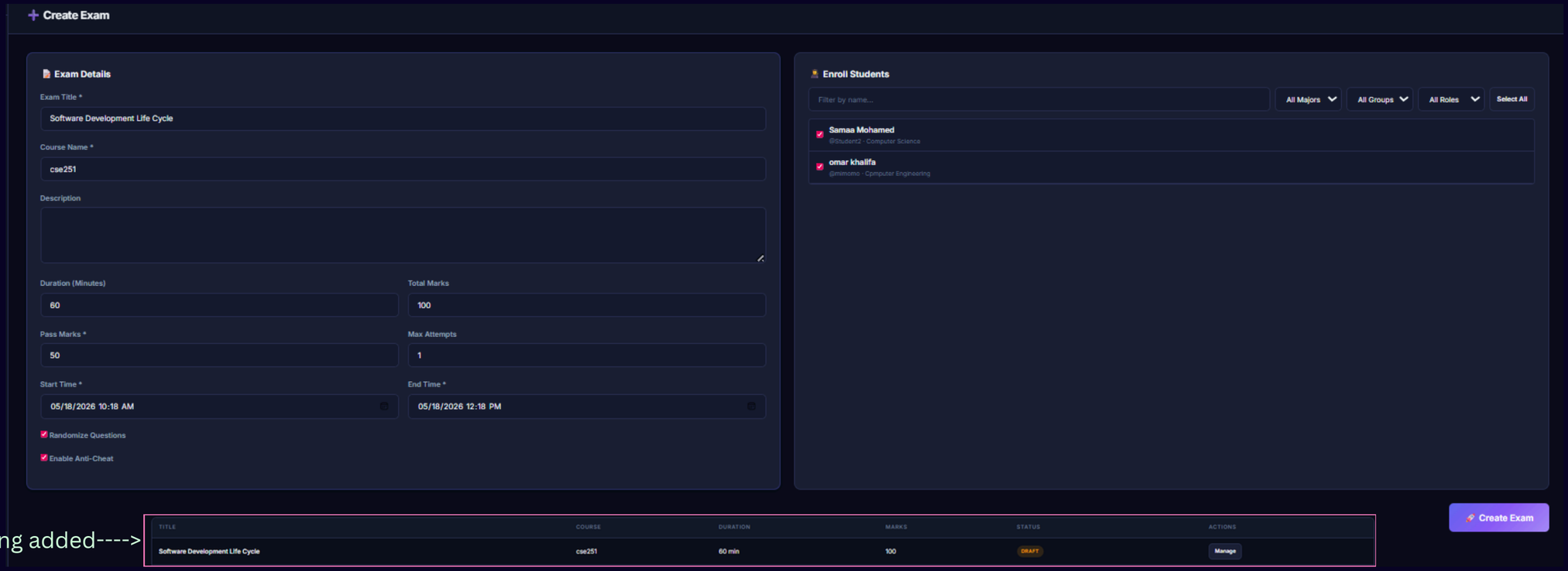
INSTRUCTOR DASHBOARD



EXAMS



CREATING AN EXAM



MANAGING EXAM QUESTIONS

Manage Questions

Software Development Life Cycle — Questions (0)

Add Question

Question Text *

SO FEOFEOFMFE FEFOFEEO FOEMFMO OFOMEF OFMEFMO?

Type

Multiple Choice

Marks

5

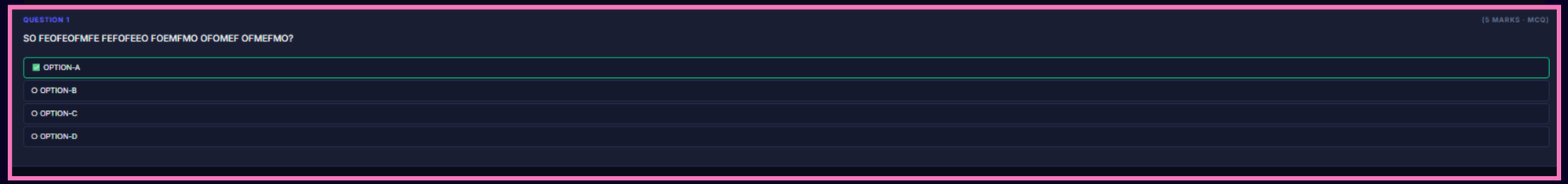
Options (one per line, for MCQ/TF)

OPTION-A
OPTION-B
OPTION-C
OPTION-D

Correct Answer (Leave empty for Open Ended)

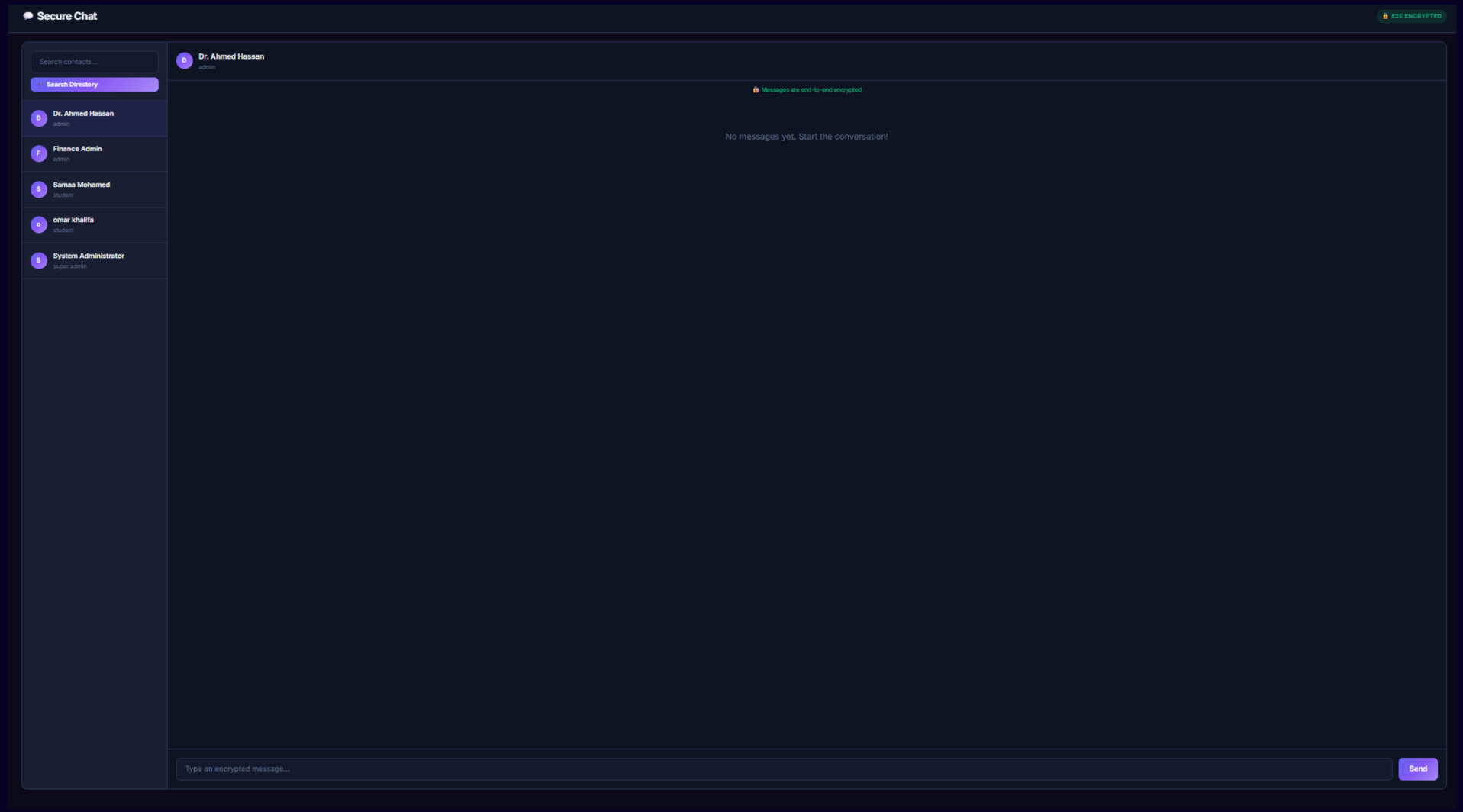
OPTION-A

Add Question



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SECURE CHAT

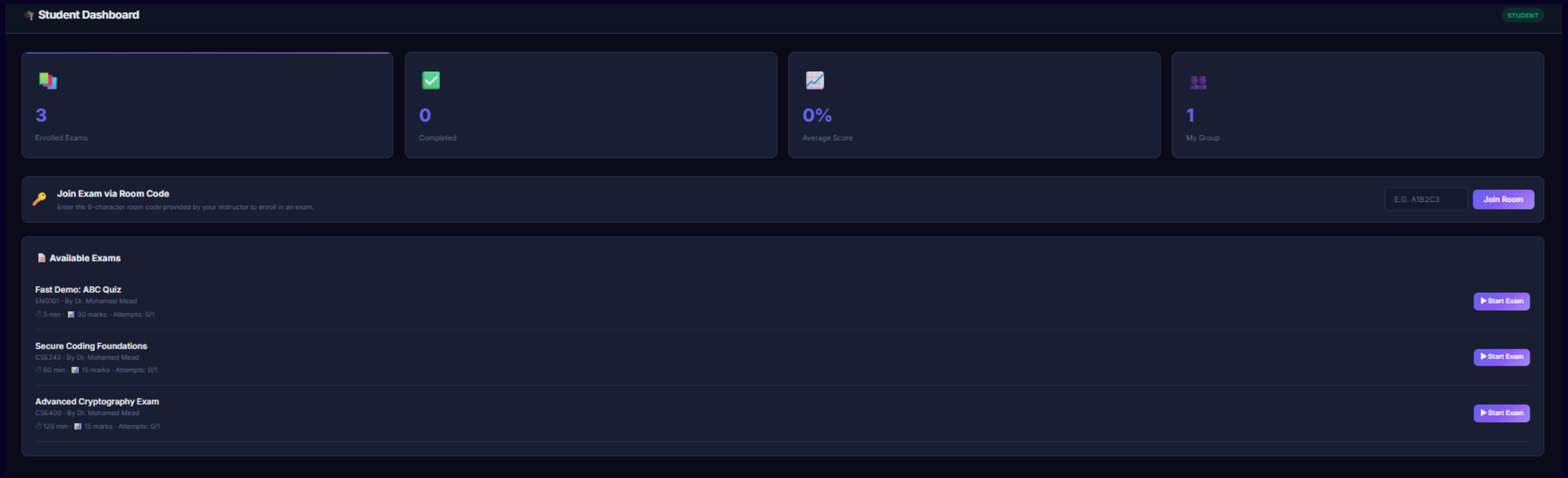


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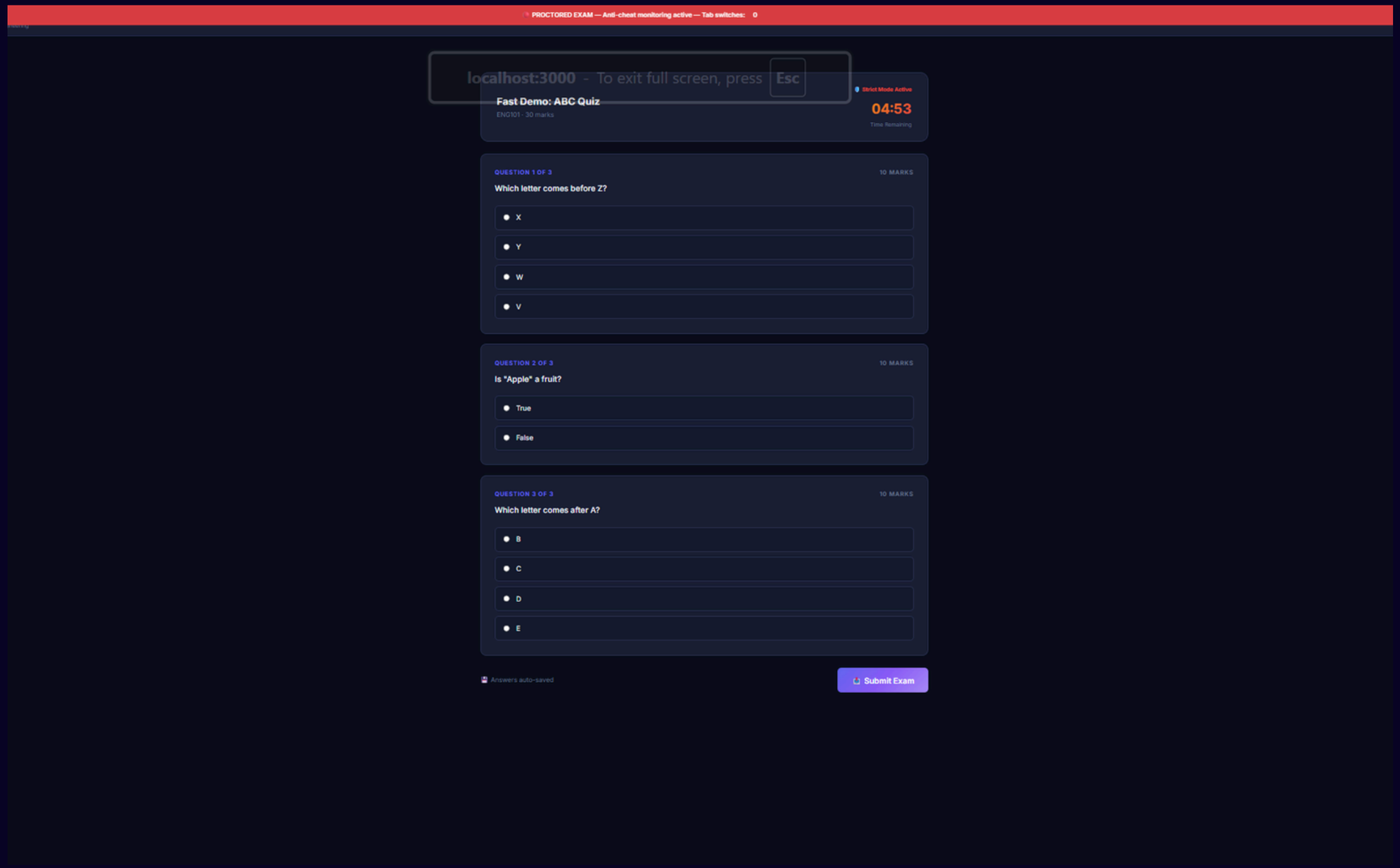
STUDENT INTERFACE

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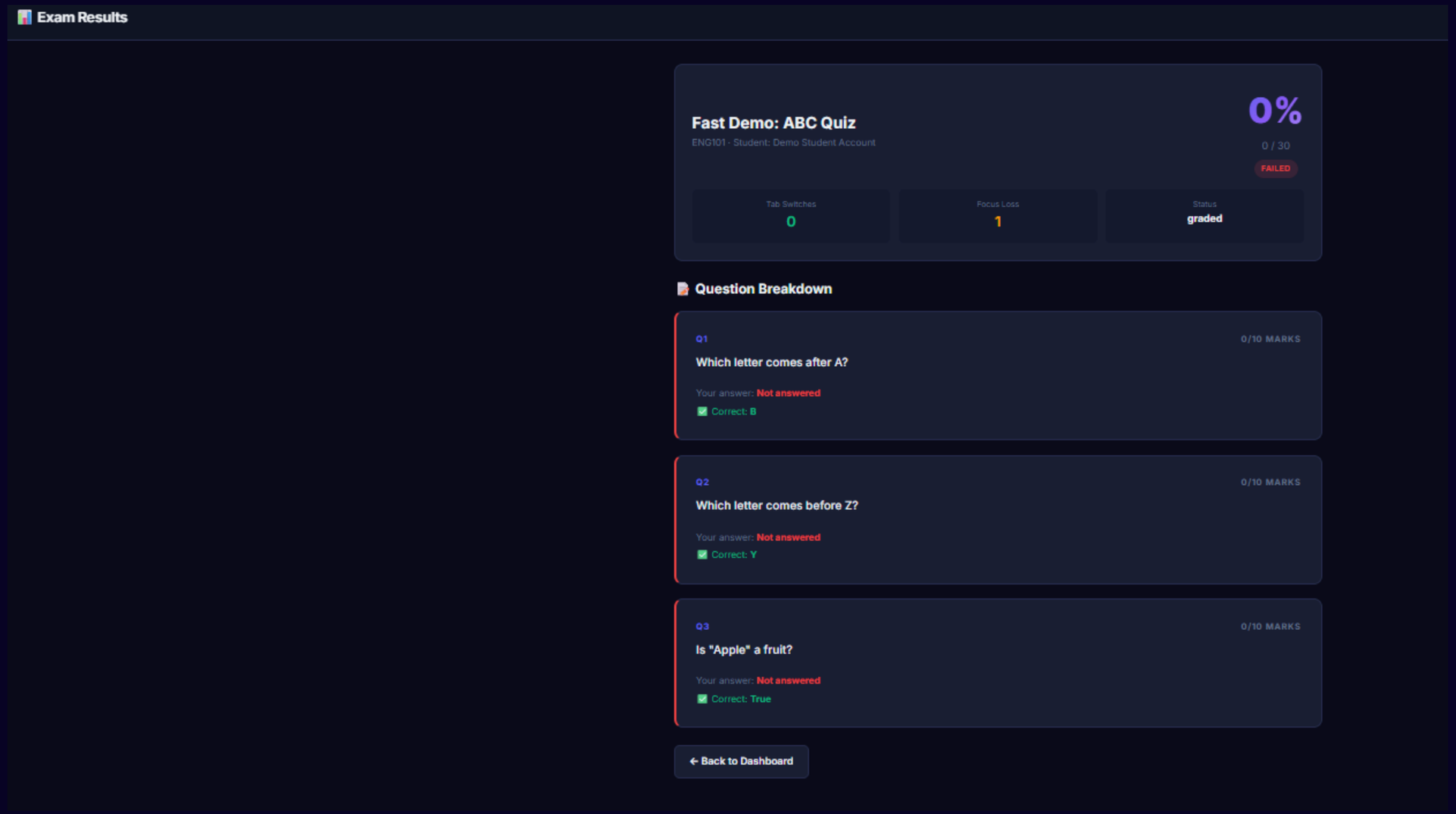
STUDENT DASHBOARD



TAKNING AN EXAM

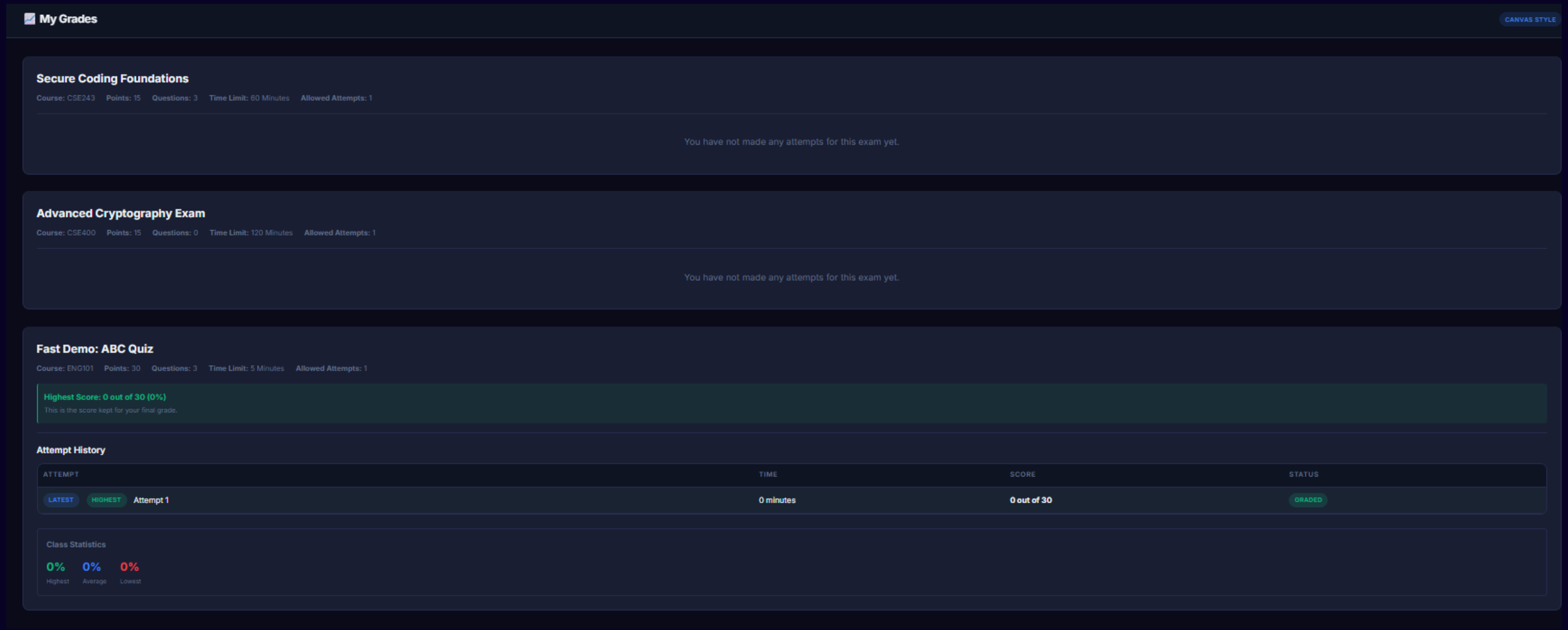


EXAM RESULTS



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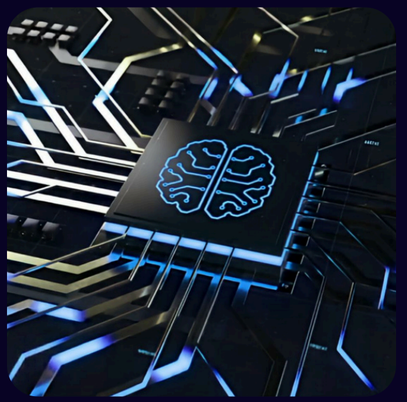
MY GRADES



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TEAM 3



THANK YOU